SHRUTI JAIN

+1-412-589-5683 | Pittsburgh, PA | spjain@andrew.cmu.edu | linkedin | personal website

EDUCATION

Carnegie Mellon University | Master of Science in Computer Vision — Aug 2025 - (Expected) Dec 2026
Coursework: Visual Learning and Recognition, Advanced Computer Vision, Introduction to Machine Learning
CGPA : **4.11/4**

Manipal Institute of Technology | Bachelor of Technology in Information Technology — 2019-2023
CGPA : **8.71/10**

SKILLS

Programming Languages/Tools: Python, C++, Java, Docker, Kubernetes, SQL
ML and CV Frameworks/Libraries: PyTorch, OpenCV, ROS, CUDA, scikit-learn, TensorFlow, Matplotlib
AI Topics: Computer Vision, Transformers, VLMs, AI Agents, Generative Models, Object Detection, Image Segmentation, Zero-shot Learning, Semi-supervised Learning, Domain Adaptation, AI evaluation

RELEVANT EXPERIENCE

FlowState AI | ***Computer Vision Research Intern*** — Jan 2026 - April 2026

- [Slides] Architected and implemented an end-to-end **AI video-editing system** that converts long-form footage into prompt-driven, story-based intelligently re-framed (ex: from 16:9 to 9:16) content.
- Built an end-to-end **multimodal computational cinematography pipeline** for story planning, shot detection, subject grounding and tracking, automated reframing, workflow orchestration, and self-evaluation.

AIRLab - Carnegie Mellon University | ***Research Assistant*** — Jan 2026 - April 2026

- [Project Website] Developing a multimodal medical triaging pipeline for the DARPA Triage Challenge using RGB and thermal UAV imagery to predict 50+ hierarchical medical labels, under the guidance of **Prof. Sebastian Scherer**.
- Built a two-stage system combining conditional diffusion-based RGB to IR generation and VLM fine-tuning with LoRA/GRPO optimization to improve triage reasoning under limited paired RGB–IR data.

Carnegie Mellon University, USA | ***Research Assistant*** — Aug 2025 – Dec 2025

- Worked under **Prof. Deva Ramanan** on developing DetPO, a gradient-free black-box prompt optimization framework achieving up to 9.7% improvement over prior approaches on Roboflow20-VL and LVIS.

Palo Alto Networks | ***Software Developer*** — Oct 2023 - April 2025

- Worked on features and APIs of the microservices (**JAVA Spring Boot**) for ingesting events from tenant's cloud accounts (**AWS, Azure, GCP**) for near real-time threat detection using **Artificial Intelligence**.
- Integrated a **scheduler-worker service** that schedules jobs to poll cloud account for events with a **job management framework** based on **SQS Queue** and **Redis cache**.

Robotics Research Centre (RRC) - IIIT Hyderabad | ***Research Assistant*** — June 2023 – Sept 2023

- Achieved **14.4%** average decrease in the mean absolute scores across 3 datasets as compared to the SOTA for **Image Segmentation in self driving cars** in adverse weather conditions.
- Implemented a **differentiable visual and latent prompting mechanism** to finetune on **Segment Anything Model** via **adaptors** through learnable image pre-processing blocks.

RELEVANT PROJECTS

- **Gated Multimodal Web Navigation Agents** - [Poster] A gated multimodal web navigation framework combining MindAct and Qwen2-VL on the Multimodal Mind2Web benchmark, improving robustness to adversarial popup attacks.
- **Editing 3D Gaussian Splats** - [Slides] Edits **3D Gaussian Splats** based on text prompts by using **Diffusion** based 2D Image Editing techniques.
- **Flash Attention for ViT** - [Poster] Implemented FlashAttention-1 and 2 in Minitorch as custom CUDA autograd ops with tiled SRAM compute and online softmax.
- **Real-time Failed 3D Print Detector** - Detects failed 3D prints in real time using **OpenCV** and anomaly detection.
- **AUVSI SUAS Competition [link]** - Achieved Autonomous detection and classification of shapes (**55% accuracy**), letters (**75% accuracy**) and their colours from a rescue drone using **OpenCV and Machine learning**.

SELECTED PUBLICATIONS

- DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection [Paper] [Website] [Code] **Published in ECCV 2026**
- DiffPrompter: Differentiable Implicit Visual Prompts for Semantic-Segmentation in Adverse Conditions [Paper] [Project Page] [Code] ***Published in IROS 2024***
- Light Weight Character and Shape Recognition for Autonomous Drones [Paper] ***Workshop acceptance in ICML 2022***
- A Studious Approach To Semi-Supervised Learning [Paper] [Code] ***Workshop acceptance in NeurIPS 2021***